LAPORAN TUGAS

MATA KULIAH WORKSHOP DEVELOPER OPERATIONAL

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KATA PENGANTAR

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BAB 1 PEMBAHASAN

1.1 CI/CD

CI/CD adalah metode untuk mengirimkan aplikasi kepada pengguna secara terus-menerus dengan menggunakan otomatisasi. Terdiri dari dua bagian utama:

1) Continuous Integration (CI): Setiap kali developer mengubah kode, kode tersebut langsung digabungkan ke pusat penyimpanan (repository), lalu diuji secara otomatis. Tujuannya agar kesalahan (bug) ditemukan lebih cepat.
2) Continuous Delivery/Deployment (CD): Setelah lulus uji, kode tersebut otomatis disiapkan untuk dirilis (Delivery) atau bahkan langsung diluncurkan ke server produksi (Deployment) tanpa campur tangan manusia.

1.2 Jenkins

Jenkins adalah tools *open-source* yang menjalankan proses CI/CD tersebut.

Jenkins bertugas :

1) Mengambil kode terbaru dari GitHub/GitLab.
2) Menjalankan perintah untuk membangun (*build*) aplikasi.
3) Menjalankan tes otomatis.
4) Mengirimkan hasilnya ke server tujuan

1.3 Jenkins Installation (WSL)

a. Masuk ke WSL

```
devi_zahra_safrida@LAPTOP-PLBPPBOU:~$ wsl
WSL
Wsman Shell commandLine, version 0.2.1

USAGE: wsl COMMAND [PARAMS...]

COMMANDS:
identify  - WS-Identify
enum      - WS-Enumerate
get       - WS-Get
put       - WS-Put
invoke    - WS-Invoke
xclean    - Delete all files generated by this tool set
xcred     - Create or display credential file
xcert     - Get server certificate (saved to <IPADDRESS>.crt)

PARAMS specification is specific to a COMMAND.

Output will be saved to ./response.xml. If you want to run parallel
executions in the same directory, define RTFILEPREFIX in the environment.
Doing so may significantly increase files generated.

Requires: curl, xmllint, GNU core utilities.
Optional: xsltproc for output formatting, gpg for encrypted credential.
Optional: wget as alternate for curl when not available.
devi_zahra_safrida@LAPTOP-PLBPPBOU:~$ sudo su
root@LAPTOP-PLBPPBOU:/home/devi_zahra_safrida# cd ~
root@LAPTOP-PLBPPBOU:~# cd /home/devi_zahra_safrida
```

Buka terminal dan masuk ke mode root/user utama.

b. Install Java JDK

Java JDK dibutuhkan untuk menjalankan Jenkins, disini saya menggunakan versi Java JDK 17.

```
root@LAPTOP-PLBPPBOU:/home/devi_zahra_safrida# sudo apt update
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Ign:3 https://pkg.jenkins.io/debian-stable binary/ InRelease
Hit:4 https://pkg.jenkins.io/debian-stable binary/ Release
Hit:6 https://download.docker.com/linux/ubuntu noble InRelease
Get:7 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:8 https://apt.releases.hashicorp.com noble InRelease [12.9 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Get:11 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [74.1 kB]
Get:12 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [177 kB]
Get:14 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [212 B]
Get:15 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [386 kB]
Get:16 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:17 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:18 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7312 B]
Get:19 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [10.5 kB]
Get:20 http://archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [216 B]
Get:21 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Fetched 1070 kB in 23s (46.7 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
19 packages can be upgraded. Run 'apt list --upgradable' to see them.
W: https://apt.releases.hashicorp.com/dists/noble/InRelease: Key is stored in legacy trusted.gpg keyring (/etc/apt/trusted.gpg),
see the DEPRECATION section in apt-key(8) for details.
root@LAPTOP-PLBPPBOU:/home/devi_zahra_safrida# sudo apt install openjdk-17-jdk -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
openjdk-17-jdk is already the newest version (17.0.18+8-1~24.04.1).
0 upgraded, 0 newly installed, 0 to remove and 19 not upgraded.
root@LAPTOP-PLBPPBOU:/home/devi_zahra_safrida# java -version
openjdk version "17.0.18" 2026-01-20
OpenJDK Runtime Environment (build 17.0.18+8-Ubuntu-124.04.1)
OpenJDK 64-Bit Server VM (build 17.0.18+8-Ubuntu-124.04.1, mixed mode, sharing)
```

c. Tambahkan Jenkins Repository Key

```
root@LAPTOP-PLBPPBOU:/home/devi_zahra_safrida# sudo wget -O /etc/apt/keyrings/jenkins-keyring.asc \
https://pkg.jenkins.io/debian-stable/jenkins.io-2026.key
--2026-03-10 18:28:40--  https://pkg.jenkins.io/debian-stable/jenkins.io-2026.key
Resolving pkg.jenkins.io (pkg.jenkins.io)... 146.75.46.133, 2a04:4e42:7b::645
Connecting to pkg.jenkins.io (pkg.jenkins.io)|146.75.46.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1680 (1.6K) [application/octet-stream]
Saving to: '/etc/apt/keyrings/jenkins-keyring.asc'

/etc/apt/keyrings/je 100%[===================>]   1.64K  --.-KB/s    in 0s

2026-03-10 18:28:41 (31.9 MB/s) - '/etc/apt/keyrings/jenkins-keyring.asc' saved [1680/1680]
```

d. Tambahkan Jenkins Repository

```
root@LAPTOP-PLBPPBOU:/home/devi_zahra_safrida# echo "deb [signed-by=/etc/apt/keyrings/jenkins-keyring.asc]" \
https://pkg.jenkins.io/debian-stable binary/ | sudo tee \
/etc/apt/sources.list.d/jenkins.list > /dev/null
```

e. Install Jenkins

```
root@LAPTOP-PLBPPBOU:/home/devi_zahra_safrida# sudo apt update
Ign:1 https://pkg.jenkins.io/debian-stable binary/ InRelease
Hit:2 https://pkg.jenkins.io/debian-stable binary/ Release
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:5 https://download.docker.com/linux/ubuntu noble InRelease
Hit:6 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:7 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:8 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Get:9 https://apt.releases.hashicorp.com noble InRelease [12.9 kB]
Fetched 12.9 kB in 2s (5266 B/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
19 packages can be upgraded. Run 'apt list --upgradable' to see them.
W: https://apt.releases.hashicorp.com/dists/noble/InRelease: Key is stored in legacy trusted.gpg keyring (/et
t/trusted.gpg), see the DEPRECATION section in apt-key(8) for details.
root@LAPTOP-PLBPPBOU:/home/devi_zahra_safrida# sudo apt install jenkins -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
jenkins is already the newest version (2.541.2).
0 upgraded, 0 newly installed, 0 to remove and 19 not upgraded.
root@LAPTOP-PLBPPBOU:/home/devi_zahra_safrida# sudo systemctl start jenkins
root@LAPTOP-PLBPPBOU:/home/devi_zahra_safrida# sudo systemctl enable jenkins
Synchronizing state of jenkins.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable jenkins
```

```
root@LAPTOP-PLBPPBOU:/home/devi_zahra_safrida# sudo systemctl start jenkins
root@LAPTOP-PLBPPBOU:/home/devi_zahra_safrida# sudo systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
     Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: en>
     Active: active (running) since Tue 2026-03-10 13:56:16 WIB; 2min 56s ago
   Main PID: 180 (java)
      Tasks: 67 (limit: 9354)
     Memory: 898.4M (peak: 906.3M)
        CPU: 1min 372ms
     CGroup: /system.slice/jenkins.service
             └─180 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/je>

Mar 10 13:55:55 LAPTOP-PLBPPBOU jenkins[180]: [LF]> This may also be found at: /v>
Mar 10 13:55:55 LAPTOP-PLBPPBOU jenkins[180]: [LF]>
Mar 10 13:55:55 LAPTOP-PLBPPBOU jenkins[180]: [LF]> ****************************>
Mar 10 13:55:55 LAPTOP-PLBPPBOU jenkins[180]: [LF]> ****************************>
Mar 10 13:55:55 LAPTOP-PLBPPBOU jenkins[180]: [LF]> ****************************>
Mar 10 13:56:16 LAPTOP-PLBPPBOU jenkins[180]: 2026-03-10 06:56:16.120+0000 [id=41>
Mar 10 13:56:16 LAPTOP-PLBPPBOU jenkins[180]: 2026-03-10 06:56:16.184+0000 [id=25>
Mar 10 13:56:16 LAPTOP-PLBPPBOU systemd[1]: Started jenkins.service - Jenkins Con>
Mar 10 13:56:19 LAPTOP-PLBPPBOU jenkins[180]: 2026-03-10 06:56:19.029+0000 [id=71>
Mar 10 13:56:19 LAPTOP-PLBPPBOU jenkins[180]: 2026-03-10 06:56:19.030+0000 [id=71>
```

Perintah :

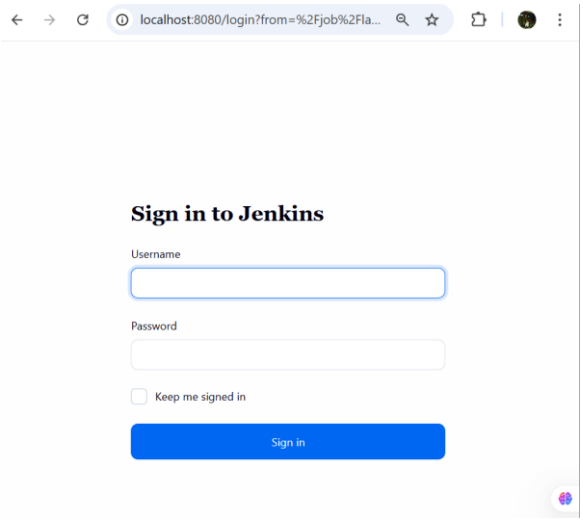
```
sudo systemctl start jenkins
sudo systemctl enable jenkins
```

Digunakan untuk menjalankan Jenkins.

f. Cek apakah Jenkins berjalan

```
root@LAPTOP-PLBPPBOU:/home/devi_zahra_safrida# sudo lsof -i :8080
COMMAND PID    USER   FD   TYPE DEVICE SIZE/OFF NODE NAME
java    186 jenkins    8u  IPv6   6832      0t0  TCP *:http-alt (LISTEN)
```

g. Buka Jenkins Dashboard



Ketika port 8080 bisa diakses melalui http://localhost:8080 berarti Jenkins berhasil dijalankan.

Kemudian ambil password awal Jenkins dengan perintah :

```
sudo cat /var/lib/jenkins/secrets/initialAdminPassword
```

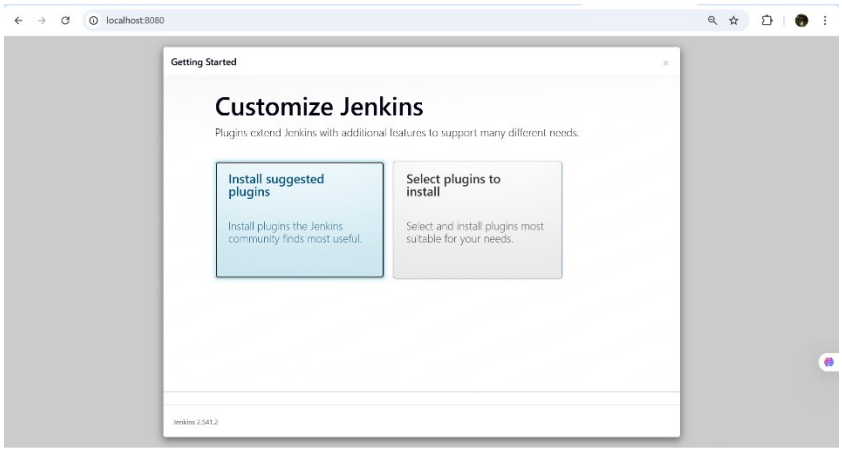
Password ini digunakan pada saat di halaman login Jenkins.

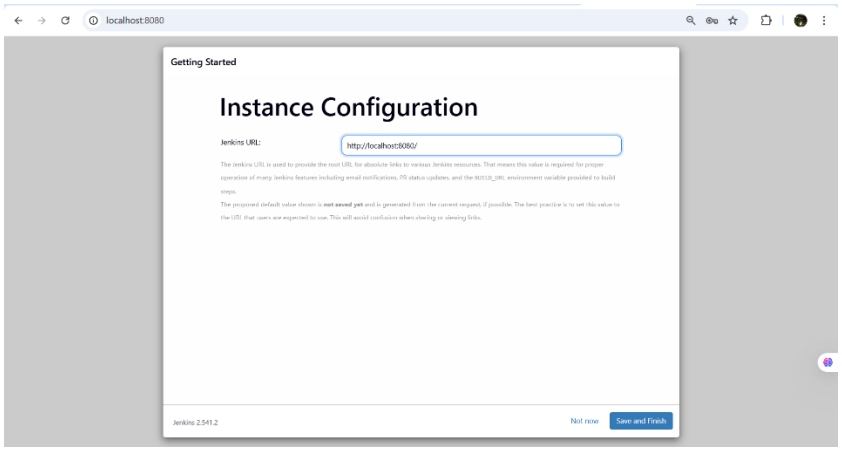
```
root@LAPTOP-PLBPPBOU:/home/devi_zahra_safrida# sudo cat /var/lib/jenkins/secrets/initialAdminPassword
16f73a41752b47c586262047aa797b15
```

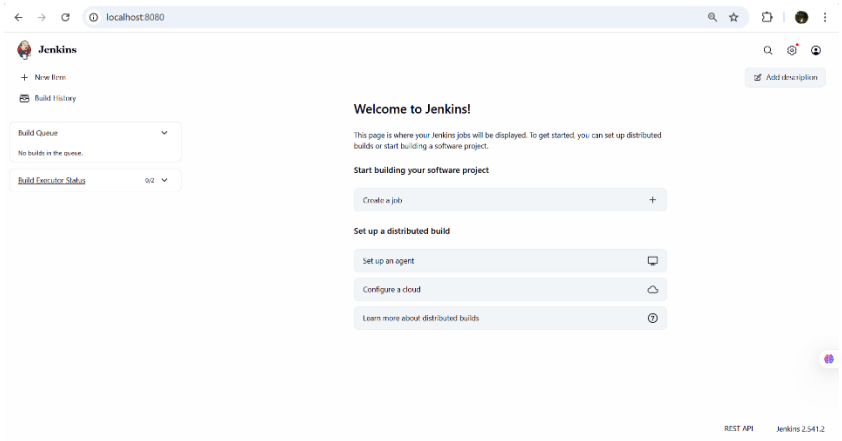
h. Install Jenkins Plugins

Pada bagian ini, pilih “Install Suggested Plugins”.

Setelah selesai, Jenkins akan masuk ke Dashboard.

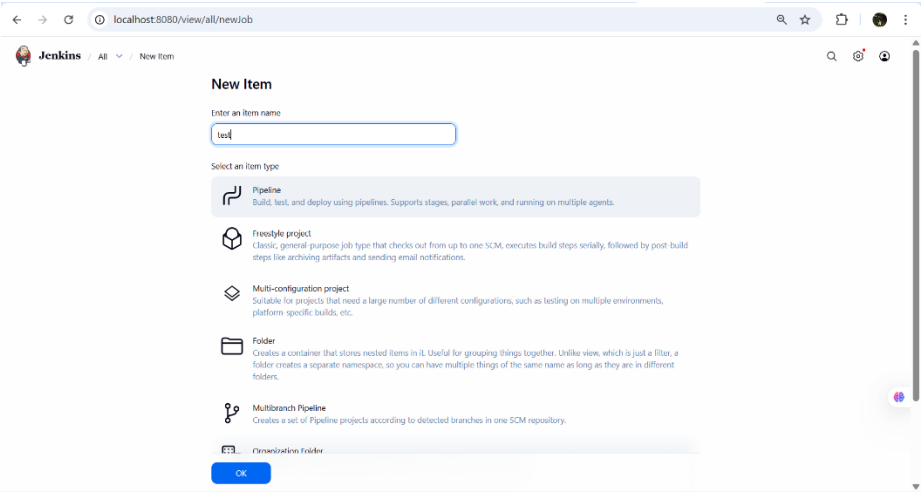




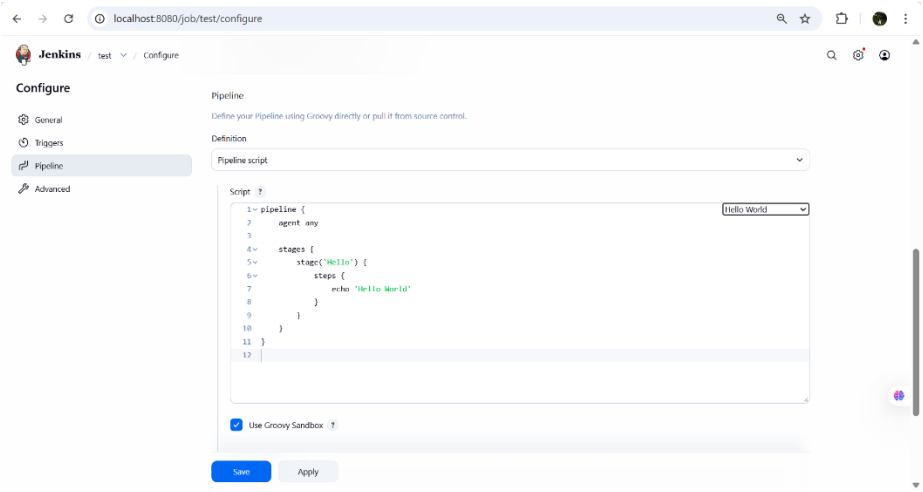


i. Test Jenkins Pipeline

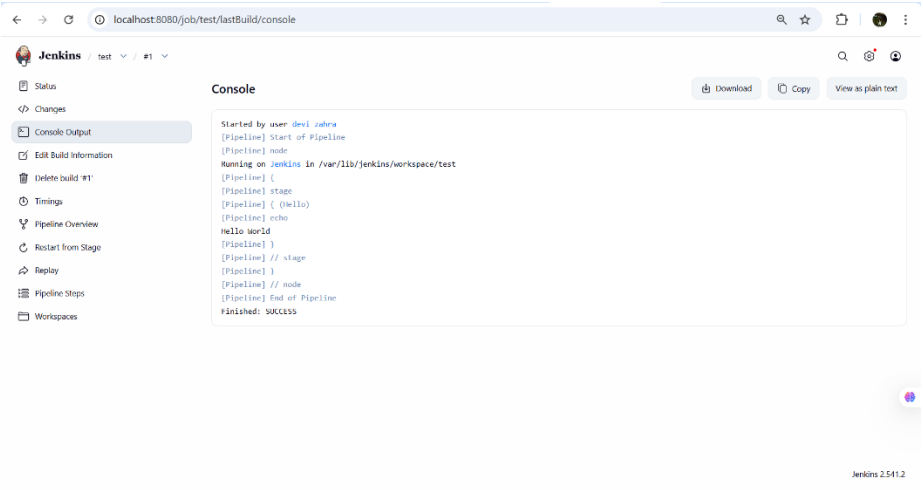
Disini kita test Pipeline dengan membuat job sederhana. Pilih ‘New Item’ dan beri nama ‘test’, dan pilih type Pipeline.



Lalu, pilih definition ‘Pipeline Script” dan pada script pilih “Hello Word”, dan save.



Kemudian, klik ‘Build Now’, lihat hasilnya di ‘Console Ouput’, apabila muncul ‘Finished: SUCCESS’, maka dinyatakan telah berhasil.



1.4 Laravel Project Setup

a. Buat Folder Project

Masuk ke cd C:\Users\USERNAME\Documents, lalu buat direktori website dan Laravel_jenkins.

```
C:\Users\DEVI ZAHRA SAFRIDA>cd Documents

C:\Users\DEVI ZAHRA SAFRIDA\Documents>mkdir website

C:\Users\DEVI ZAHRA SAFRIDA\Documents>cd website

C:\Users\DEVI ZAHRA SAFRIDA\Documents\website>mkdir laravel_jenkins
```

b. Install Laravel

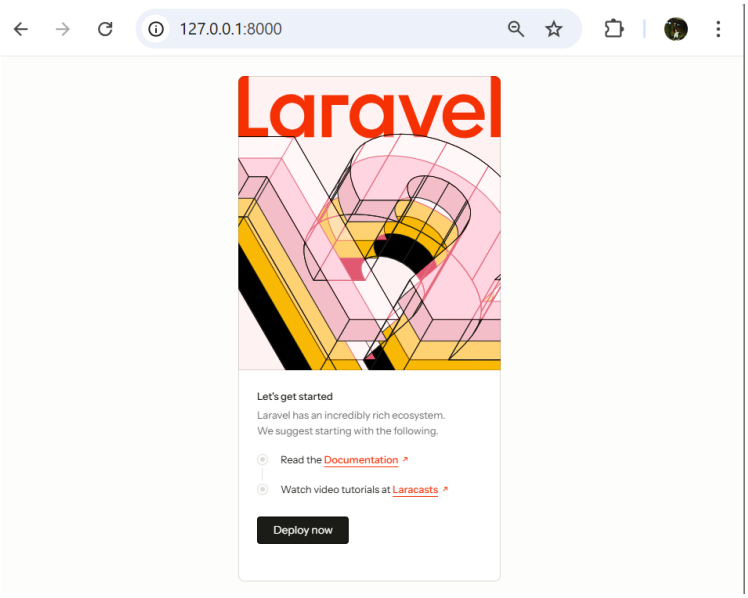
Laravel ini nantinya digunakan sebagai project yang akan dijalankan.

```
C:\Users\DEVI ZAHRA SAFRIDA\Documents\website>composer create-project laravel/laravel laravel-ci
Creating a "laravel/laravel" project at "./laravel-ci"
Installing laravel/laravel (v12.11.2)
  - Downloading laravel/laravel (v12.11.2)
  - Installing laravel/laravel (v12.11.2): Extracting archive
```

c. Setup Environment dan jalankan Laravel

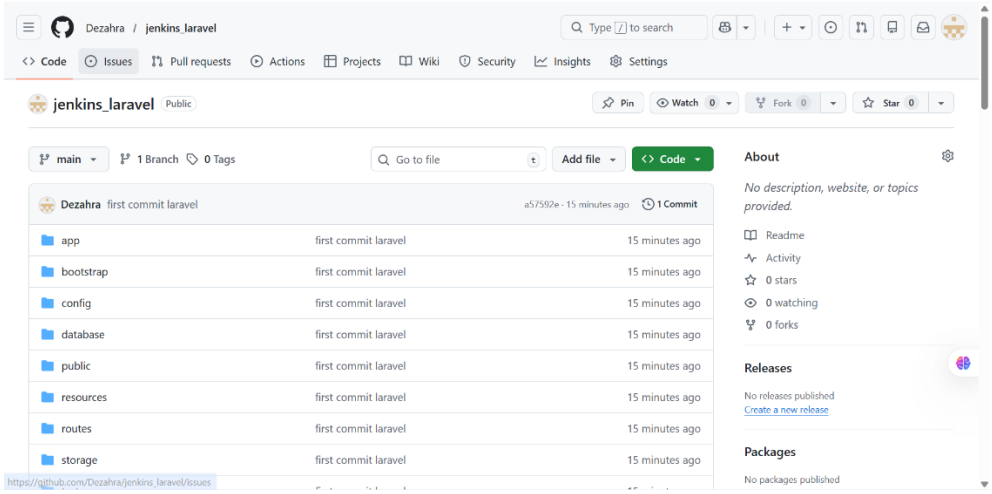
```
No security vulnerability advisories found.
> @php artisan key:generate --ansi

  INFO  Application key set successfully.

> @php -r "file_exists('database/database.sqlite') || touch('database/database.sqlite');"
> @php artisan migrate --graceful --ansi

  INFO  Preparing database.

  Creating migration table ..................................................... 10.35ms DONE

  INFO  Running migrations.

  0001_01_01_000000_create_users_table ......................................... 24.54ms DONE
  0001_01_01_000001_create_cache_table ......................................... 13.42ms DONE
  0001_01_01_000002_create_jobs_table .......................................... 18.54ms DONE


C:\Users\DEVI ZAHRA SAFRIDA\Documents\website>cd laravel-ci

C:\Users\DEVI ZAHRA SAFRIDA\Documents\website\laravel-ci>copy .env.example .env
Overwrite .env? (Yes/No/All): all
        1 file(s) copied.

C:\Users\DEVI ZAHRA SAFRIDA\Documents\website\laravel-ci>php artisan key:generate

  INFO  Application key set successfully.


C:\Users\DEVI ZAHRA SAFRIDA\Documents\website\laravel-ci>php artisan serve

  INFO  Server running on [http://127.0.0.1:8000].

  Press Ctrl+C to stop the server
```

Lalu akses di browser melalui http://127.0.0.1:8000.

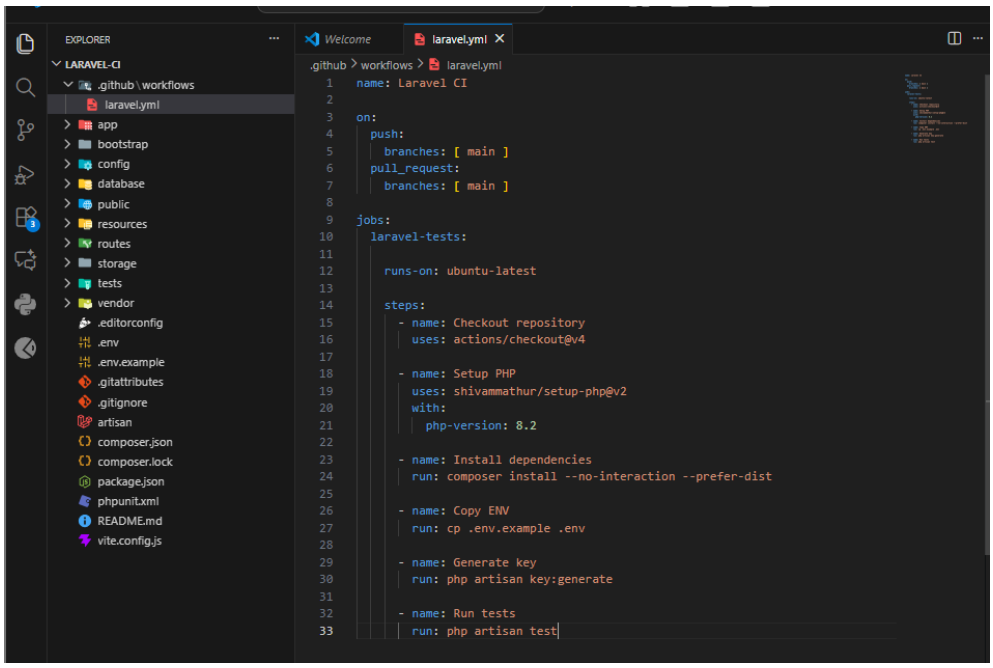


1.5 CI/CD using GitHub Actions

Setelah berhasil membuat project Laravel, selanjutnya push project ke GitHub. Create repository dengan nama ‘jenkins_laravel’.



Buka project Laravel bisa melalui VSCode. Lalu buat folder workflow ‘.github/workflows’, lalu buat file didalamnya ‘laravel.yml’. Setelah itu, push lagi ke github.



1.6 CI/CD using Jenkins

a. Install Required Jenkins Plugins

Masuk ke Manage Jenkins, lalu pilih Plugins. Install beberapa plugin berikut :

- Git plugin
- GitHub Integration
- Credentials Binding
- Pipeline
- Pipeline: Stage View

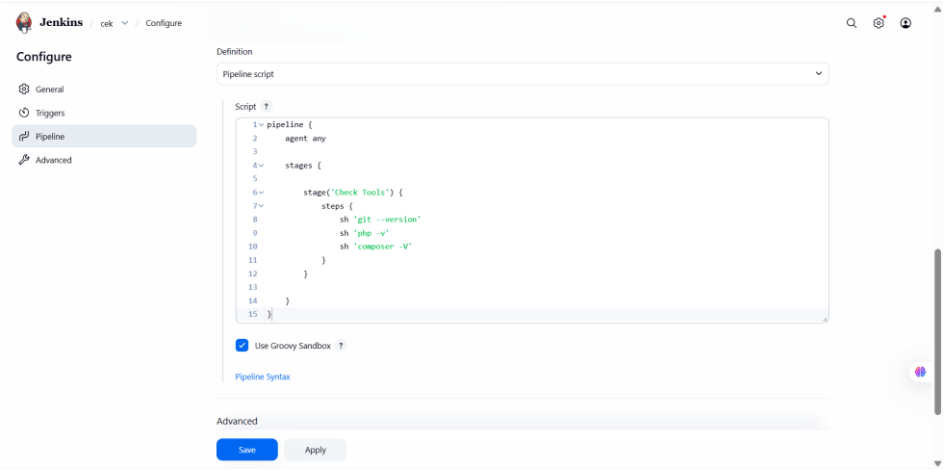
Selanjutnya, install tools php dan composer.

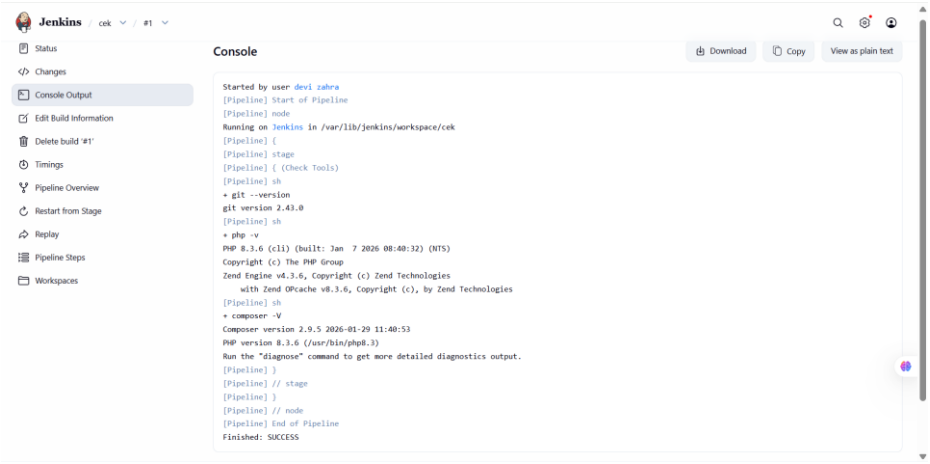
```
root@LAPTOP-PLBPPBOU:/home/devi_zahra_safrida# sudo apt install php-cli php-json php-zip wget unzip curl -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
php-cli is already the newest version (2:8.3+93ubuntu2).
php-zip is already the newest version (2:8.3+93ubuntu2).
php-zip set to manually installed.
```

```
root@LAPTOP-PLBPPBOU:/home/devi_zahra_safrida# curl -sS https://getcomposer.org/installer -o composer-setup.php
root@LAPTOP-PLBPPBOU:/home/devi_zahra_safrida# sudo php composer-setup.php --install-dir=/usr/local/bin --filenam
e=composer
All settings correct for using Composer
Downloading...

Composer (version 2.9.5) successfully installed to: /usr/local/bin/composer
Use it: php /usr/local/bin/composer
```

Cek Pipeline dengan membuat item baru. Jika berhasil, akan terlihat versi Git, PHP, dan Composer yang terinstal di WSL. Di bagian paling bawah harus tertulis:

Finished: SUCCESS





b. Install PHP & Composer on Jenkins (WSL)

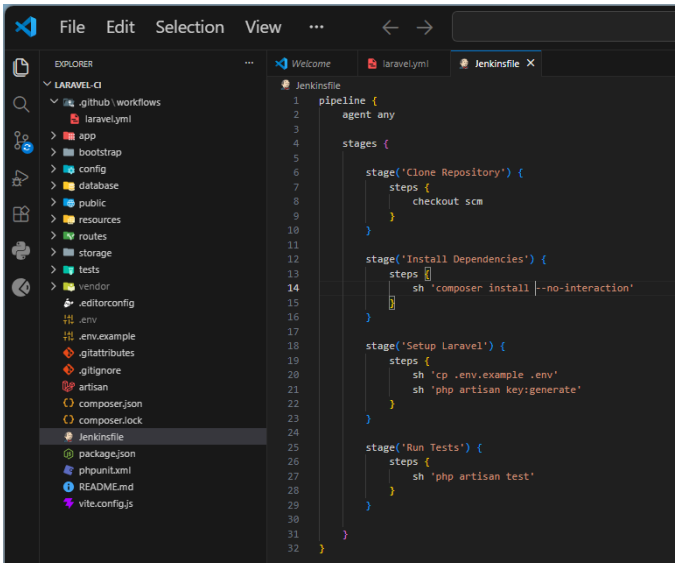
Selanjutnya, install php dan composer di WSL.

```
root@LAPTOP-PLBPPBOU:/home/devi_zahra_safrida# sudo apt install php php-cli php-mbstring php-xml php-curl unzip -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
php-cli is already the newest version (2:8.3+93ubuntu2).
unzip is already the newest version (6.0-28ubuntu4.1).
The following additional packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils libapache2-mod-php8.3 libapr1t64 libaprutil1-dbd-sqlite3
  libaprutil1-ldap libaprutil1t64 liblua5.4-0 libonig5 php8.3 php8.3-mbstring ssl-cert
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom www-browser ufw php-pear
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils libapache2-mod-php8.3 libapr1t64 libaprutil1-dbd-sqlite3
  libaprutil1-ldap libaprutil1t64 liblua5.4-0 libonig5 php php-curl php-mbstring php-xml php8.3 php8.3-mbstring
  ssl-cert
```

```
root@LAPTOP-PLBPPBOU:/home/devi_zahra_safrida# sudo apt install composer -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  jsonlint libzip4t64 php-composer-ca-bundle php-composer-class-map-generator php-composer-metadata-minifier php-composer-pcre
  php-composer-semver php-composer-spdx-licenses php-composer-xdebug-handler php-intl php-json-schema php-psr-container
  php-psr-log php-react-promise php-seld-signal-handler php-symfony-console php-symfony-deprecation-contracts
  php-symfony-filesystem php-symfony-finder php-symfony-process php-symfony-service-contracts php-symfony-string php-zip
  php8.3-intl php8.3-zip
Suggested packages:
  fossil mercurial subversion
The following NEW packages will be installed:
```

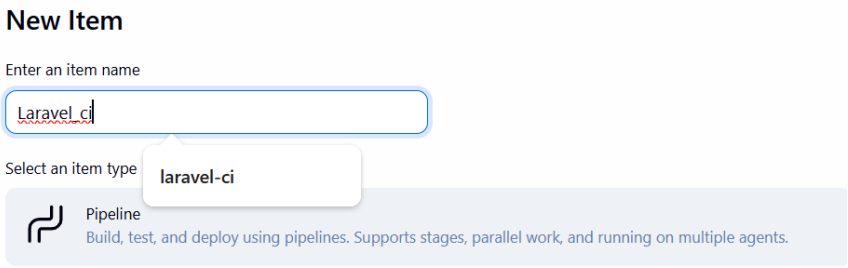
c. Jenkins Pipeline Script

Buat file di root project Laravel di VSCode dengan nama file 'Jenkinsfile'. Lalu push file ini ke GitHub.

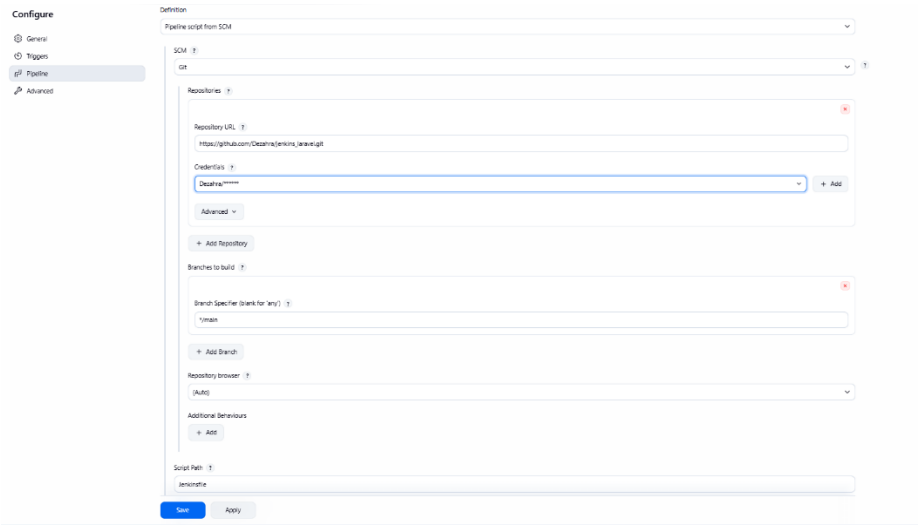


d. Jenkins Pipeline Configuration

Masuk ke halaman Jenkins, lalu pilih ‘New Item’, beri nama item ‘laravel-ci’, dan pilih tipe ‘Pipeline’.



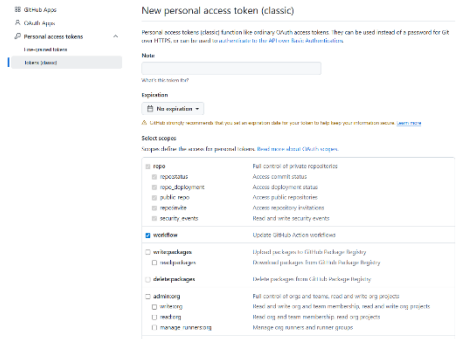
Atur configurasi Pipeline. Pada Definition pilih ‘Pipeline script from SCM’, pada SCM pilih ‘Git’, pada Repo masukkan link GitHub, dan pada Branch masukkan ‘*/main’.



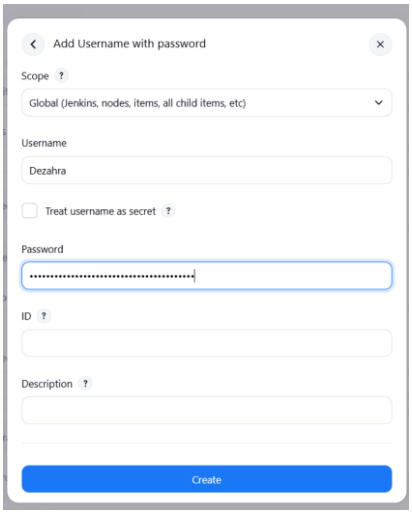
e. GitHub Credential Setup

Untuk memasukkan repository GitHub pada Pipeline Configuration diperlukan token dari GitHub. Jadi bikin tokennya dahulu.

Untuk generate GitHub Toke, masuk ke pengaturan GitHub, pilih ‘Developer settings’, pilih ‘Personal access tokens’, dan pilih ‘Tokens (classic). Centang permission repo dan workflow.



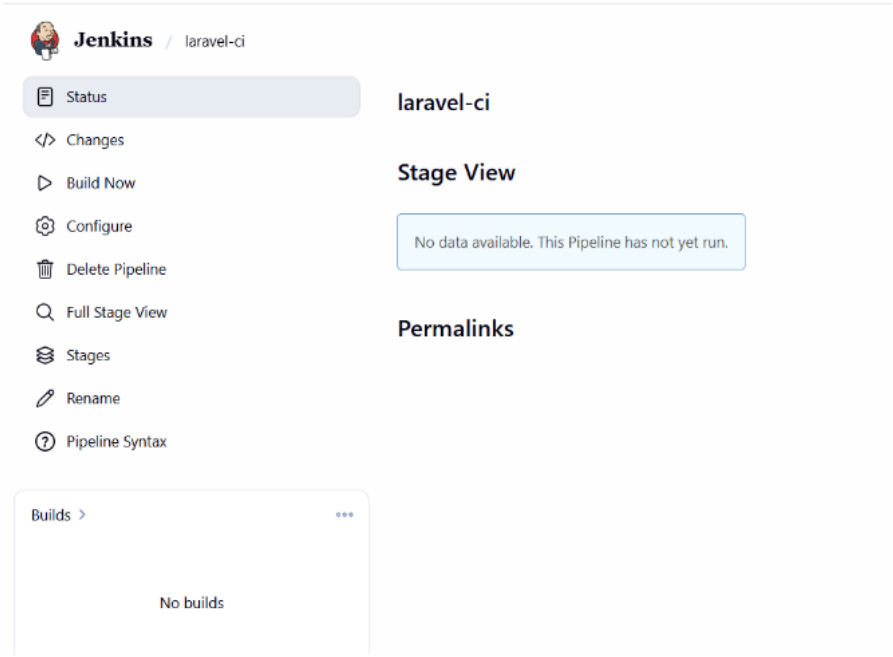
Tambah credential. Kind: Username with password ; Username: GitHub username; Password: GitHub Personal Access Token.



f. Run Jenkins Pipeline

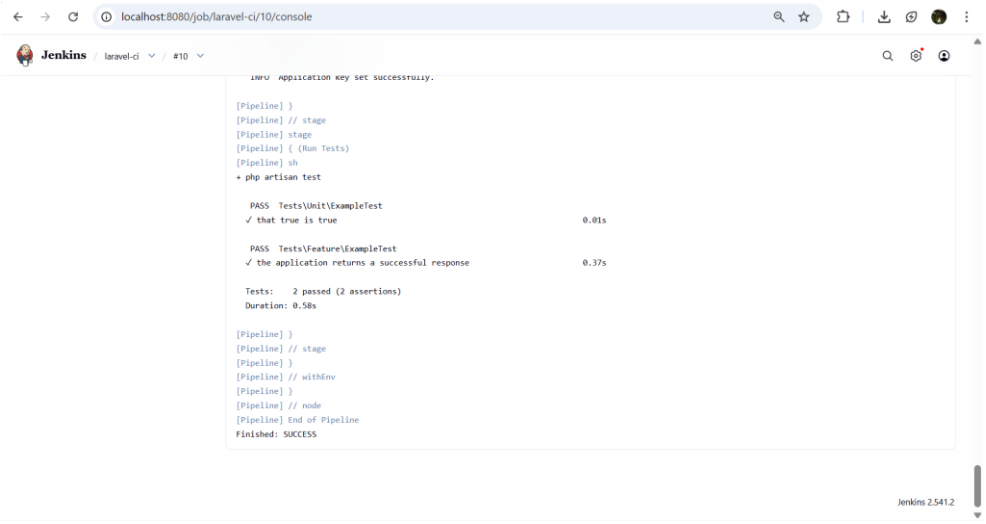
Selanjutnya yaitu menjalankan Jenkins Pipeline. Klik 'Build Now', maka akan memulai build pipeline. Pipeline akan menjalankan Clone Repository, Composer Install, Laravel Setup, dan Run Tests.

Jika berhasil, maka akan muncul Finished: SUCCESS di console.



1.7 Final CI/CD Workflow

Terlihat run Jenkins Pipeline berhasil.



Alur kerja otomatisasi (Workflow), sebagai berikut :

1. Developer Push
   Proses dimulai ketika mengirimkan perubahan kode terbaru dari komputer lokal ke GitHub menggunakan perintah git push origin main.
2. GitHub Repository
   GitHub bertindak sebagai pusat penyimpanan kode (Single Source of Truth). Setiap perubahan yang masuk akan memicu webhook untuk menjalankan tahap berikutnya.
3. GitHub Actions CI
   Sebelum kode masuk ke Jenkins, GitHub Actions melakukan pengecekan awal di awan (Cloud).
   Fungsi: Menjalankan linter atau tes cepat pada lingkungan ubuntu-latest.
   Keuntungan: Mendeteksi error sintaksis dasar sebelum membebani server Jenkins lokal Anda.
4. Jenkins Pipeline
   Setelah lolos dari GitHub Actions, Jenkins (yang berjalan di WSL) mengambil alih kendali. Jenkins akan menarik kode terbaru (Pull) dan mengeksekusi instruksi di dalam Jenkinsfile.
5. Build + Test Laravel Application
   Ini adalah tahap pembuktian di lingkungan Environment yang sesungguhnya:
   - Install: Menjalankan composer install (mengunduh vendor).
   - Setup: Mengatur .env dan key:generate.
   - Test: Menjalankan php artisan test untuk memastikan fitur aplikasi tidak rusak setelah ada kode baru.

BAB 2 PENUTUP

2.1 Kesimpulan

Implementasi ini menunjukkan bahwa adopsi metodologi CI/CD melalui integrasi GitHub Actions dan Jenkins secara signifikan mengoptimalkan efisiensi siklus pengembangan aplikasi Laravel. Pemanfaatan WSL sebagai basis infrastruktur lokal terbukti andal dalam menjembatani kebutuhan lingkungan yang fleksibel namun tetap sinkron dengan repositori berbasis cloud. Keberhasilan alur kerja ini menegaskan bahwa otomatisasi bukan sekadar alat bantu, melainkan fondasi penting untuk menjaga integritas kode melalui tahapan testing dan build yang konsisten. Secara teknis, konfigurasi dependensi yang presisi pada level sistem operasi menjadi kunci utama dalam meminimalisir anomali saat eksekusi pipeline, sehingga produk akhir yang dihasilkan memiliki reliabilitas yang lebih tinggi dan siap untuk diimplementasikan ke tahap produksi.